

二. 广义相对论基础

1. 引力定律改写

引力普适性: 1 引力场物体普遍受力

2 无外力物体同轨迹 ($\frac{m_{\text{引}}}{m_{\text{惯}}} = C$) (等效原理)



假设: 引力为时空弯曲 (4维时空的几何效应)

2. 广相基本假设与广义协变性

假设: 1. 引力为时空弯曲 ($\mathbb{R}^4, \eta_{ab}$) $\rightarrow$ (M, g_{ab})

2. 自由质点世界线为测地线 $F^a = U^b \nabla_b P^a = 0$ ($U^b \nabla_b U^a = 0$ 为测地线)

3. 时空弯曲受质量影响

(精义) 广义协变性: 只有度规与派生量可作为背景几何出现 (主客体之别, 舞台)

最小替换法则: $\eta_{ab}, \partial_a \rightarrow g_{ab}, \nabla_a$ (称为与度规适应导数)

$$A^a := U^b \nabla_b U^a \dots \dots \text{电磁力 } F := q F_b U^b$$

$$\text{麦: } \nabla_a F_{bc} = 0, \nabla^a F_{ab} = -4\pi J_b \rightarrow T_{ab} = \frac{1}{4\pi} (F_{ac} F_b^c - \frac{1}{4} g_{ab} F_{cd} F^{cd})$$

3. 转动

去除 Z^a 对 $\frac{D}{d\tau}$ 的影响

费米-沃克导数 $\frac{D_F}{d\tau}$: $\mathcal{F}_a(k, l) \rightarrow \mathcal{F}_a(k, l)$

Fermi-Walker derivative

1) 线性 2) 莱布尼兹率 3) 与偏导可交换

4) 对标量: $\frac{D_F}{d\tau} = \frac{d}{d\tau}$

5) 对(1,0)张量: $\frac{D_F v^a}{d\tau} = \frac{D v^a}{d\tau} + (A^a Z^b - A^b Z^a) v_b$ (自旋)

协变导数 $Z^b \nabla_b v^a$ (Z, A 为 $G(c)$ 上的速与加速) $\rightarrow$ 可推广到 (k, l) 张量

$$1) G(c) \text{ 为 Minkowski} \rightarrow \frac{D_F}{d\tau} v^a = \frac{D v^a}{d\tau}$$

$$2) \frac{D_F}{d\tau} Z^a = 0$$

$$3) W^a \text{ 为空间矢 } (W^a Z_a = 0) \rightarrow \frac{D_F}{d\tau} W^a = h^a_b \frac{D W^b}{d\tau} \quad (h^a_b = \eta^a_b + Z^a Z_b)$$

$$4) \frac{D_F g^{ab}}{d\tau} = 0 \quad \text{pf: } \frac{D_F (g^{ab} v_a v_b)}{d\tau} = \frac{D_F (v^a u_a)}{d\tau} = v_a \frac{D v^a}{d\tau} + u_a \frac{D v^a}{d\tau} + 4 A^a Z^b \eta_{ab} v_b$$

$$= g^{ab} v_a \frac{D v^b}{d\tau} + g^{ab} u_a \frac{D v^b}{d\tau} \rightarrow \frac{D_F g^{ab}}{d\tau} = 0$$

费移: $\frac{D_F v^a}{d\tau} = 0$ 沿测地线平移为费移; Z^a 移动为费移; 费移/保长度 $\rightarrow$ 保内积 对有旋转矢, $\frac{D_F v^a}{d\tau} = \hat{\Omega}_{ab} v^a$

Ω_{ab} 规范自由性: 矢量绕自身转动无影响

若 $\Lambda_{ab} v^a = 0 \rightarrow \Omega'_{ab} = \Omega_{ab} + \Lambda_{ab}$

转动: $\frac{d\vec{w}(t)}{dt} = \vec{\omega} \times \vec{w} \Rightarrow \frac{dw^i(t)}{dt} = \epsilon^i_{jk} \omega^j w^k$

$\downarrow$ 诱导到空间上

$w_b = w^a \cdot h_{ab}$

角速度2-形式: $\Omega_{ab} \equiv (\star \omega)_{ab} = \omega^c \epsilon_{cab}$

$\frac{dw^i}{dt} = -\Omega^i_j w^j$; $\frac{Dv^a}{d\tau} = -\Omega^{ab} v^b \rightarrow v^b$ 在 $G_1(\tau)$ 上以 Ω^{ab} 时空转动

Claim: v^a, u^a 同转 $\rightarrow v^a u_a$ 在 $G_1(\tau)$ 为常数

Pf: $\frac{D}{d\tau}(v^a u_a) = u_a \frac{D}{d\tau} v^a + v_a \frac{D}{d\tau} u^a = -(u_a \Omega^{ab} v_b + u_b \Omega^{ab} v_a) = 0$

在时空中, 转动角速度 $\tilde{\Omega}_{ab} = A_a \wedge Z_b$

由时空弯曲产生的伪转动

在 Z_b, A_a 面转, 故 $\tilde{\Omega}_{ij} = 0$ (无纯空间项)

pf: 1) $\frac{Dv^a}{d\tau} - \frac{D_F v^a}{d\tau} = -(A^a Z^b - A^b Z^a) \cdot v_b = -\tilde{\Omega}^{ab} v_b$

2) $-\tilde{\Omega}^{ab} Z_b = -(A^a Z^b - A^b Z^a) Z_b = A^a = \frac{DZ^a}{d\tau}$



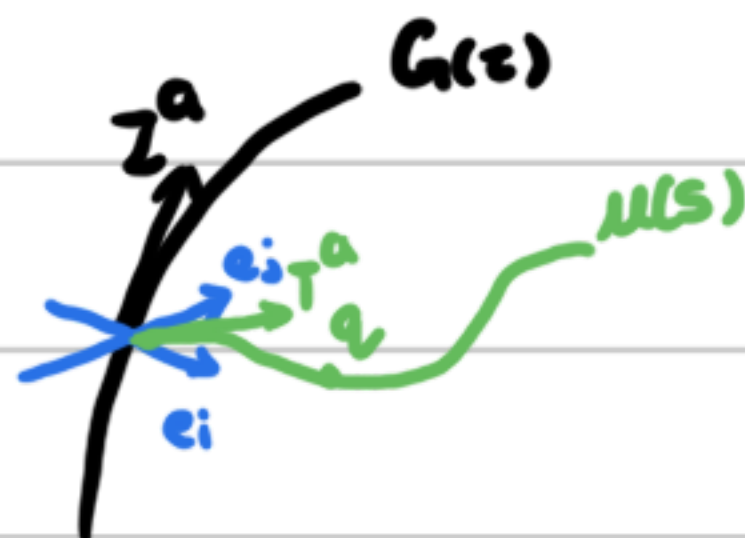
若对 w^a 为空间矢转动 Ω_{ab}

则纯空间转动 $\hat{\Omega}_{ab} = \Omega_{ab} - \tilde{\Omega}_{ab} \rightarrow \hat{\Omega}_{ab} = \epsilon_{abc} \omega^c \rightarrow \frac{D_F v^a}{d\tau} = -\epsilon^a_{bc} v^b \omega^c$

标架
证明选读

4. 固有坐标系 (4标架 $\xrightarrow{\text{延拓}}$ 4坐标系)

proper coordinate system



对于观者: $G(t), \hat{A}^a, Z^a, w^a$, 3标架场 $\{(e_i)^a\}$

固有坐标: 如图: q 在 P 邻域, $\mu(s)$ 为类空测地线, q 为 S_P (s 为线长)

q 坐标: $t(q) := T_q, x^i(q) = S_q \cdot w^i$ T_P 为 P 固有时, w^i 为 T^a 在 $(e_i)^a$ 上的分量

Claim: 固有坐标系 $g_{\mu\nu|P} = \eta_{\mu\nu}$ pf: 以 $(e_i)^a$ 为 $T^a, w^i = 1, w^j = 0 \rightarrow t = T_P, x^i = S_q \cdot w^i = 0 \rightarrow (\frac{\partial}{\partial x^i})|_P = (e_i)^a|_P, (\frac{\partial}{\partial t})|_P = Z^a|_P \rightarrow g_{ab}|_P$ 分量 $g_{\mu\nu}|_P = \eta_{\mu\nu}$ 对其它 T^a 同样, done.

$\downarrow$ $\{t, x^i\}$ 观测: 取点至少一段世界线 L 在 G 的固有坐标系内

可观测

$$\Gamma^0_{..} = \Gamma^i_{..} = 0$$

3速 $u^a := \frac{dx^a(t)}{dt} \cdot (\frac{\partial}{\partial x^i})^a$

3加速 $a^a := \frac{d^2x^a(t)}{dt^2} \cdot (\frac{\partial}{\partial x^i})^a$

Claim: g_{ab} 的克氏符在固有坐标系形式: $\Gamma^i_{..} = \hat{A}^i, \Gamma^0_{..} = \hat{A}_i; \Gamma^i_{jo} = w^k \epsilon^i_{jk}$

观者 4 加速度 $\hat{A}^a$, 转动 $w^a \rightarrow (e_i)^b \nabla_b (e_\mu)^a = \frac{D(e_\mu)^a}{ds} = -\Omega^a_{\mu b} (e_\mu)^b$ ($\Omega_{ab} = \hat{A}_a Z_b + \epsilon_{abc} w^c$)

1) 对全时 克氏符有: $(\frac{\partial}{\partial x^i})^b \nabla_b (\frac{\partial}{\partial x^i})^a = \Gamma^a_{\mu\nu} (\frac{\partial}{\partial x^i})^\mu (\frac{\partial}{\partial x^i})^\nu \rightarrow \Gamma^a_{\mu 0} (e_0)^a = -\Omega^a_{\mu b} (e_\mu)^b = -\Omega^a_{\mu b} (e_0)^b \rightarrow \Gamma^a_{\mu 0} = -\Omega^a_{\mu 0}$ ($\mu, \nu = 0, 1, 2, 3$) (ijk 40)

2) 对 $0ij$ 项: $\mu(s)$ 为测地线 $\rightarrow x^0 = t = T_P, x^i = S_q \cdot w^i \rightarrow \frac{dx^i}{ds} = 0$

$\downarrow$ $0 = \frac{dx^0}{ds} + \Gamma^0_{\mu\nu} \frac{dx^\mu}{ds} \frac{dx^\nu}{ds} = \Gamma^0_{ij} \frac{dx^i}{ds} \frac{dx^j}{ds} \rightarrow \Gamma^0_{ij} T^i T^j = 0 \forall T^a \in W_P \rightarrow \Gamma^0_{ij} = 0$ ($ij = 1, 2, 3; \mu, \nu = 0, 1, 2, 3$)

$[\hat{A}^0 = 0, Z^0 = 1, Z_a = -1] \rightarrow \begin{cases} \Gamma^0_{00} = 0 \\ \Gamma^0_{i0} = \hat{A}_i; \Gamma^i_{00} = \hat{A}^i \\ \Gamma^i_{j0} = -w^k \epsilon_{kij} = w^k \epsilon^i_{jk} \end{cases}$

$E_{abc} = Z^d \epsilon_{dabc}$ $\rightarrow$ 与 g_{ab} 适应的 4元

$\downarrow$ $w^a, \hat{A}^a$ 由 G 观察自由质点 L :

$a^a \equiv (\frac{d^2x^a}{dt^2}) (e_i)^a \Rightarrow$

$\mu=0 \rightarrow \frac{d^2t}{ds^2} = -\Gamma^0_{\mu\nu} \frac{dx^\mu}{ds} \frac{dx^\nu}{ds} = -2\Gamma^0_{0i} \frac{dt}{ds} \frac{dx^i}{ds} \gamma^i = -2\hat{A}_i w^i \gamma^i$

对 L : $\frac{d^2x^\mu}{ds^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{ds} \frac{dx^\beta}{ds} = 0, \frac{dx^\mu}{ds} = \frac{dx^\mu}{dt} \cdot \frac{dt}{ds} \rightarrow \frac{d^2x^\mu}{ds^2} = \gamma \frac{d}{ds} (\gamma \frac{dx^\mu}{dt}) = \gamma^2 \frac{d^2x^\mu}{dt^2} + \gamma \frac{d\gamma}{ds} \frac{dx^\mu}{dt} \rightarrow \frac{d^2x^0}{ds^2} = \gamma \frac{d\gamma}{dt}$

$\gamma^2 \frac{d^2x^i}{ds^2} + \gamma \frac{d\gamma}{ds} \frac{dx^i}{dt} + \Gamma^i_{0j} \gamma^2 \frac{dx^0}{ds} \frac{dx^j}{ds} = 0, a^i = \frac{d^2x^i}{dt^2} = -\frac{1}{\gamma} u^i \frac{d\gamma}{dt} - (\Gamma^i_{00} + 2\Gamma^i_{0j} u^j + \Gamma^i_{jk} u^j u^k)$

$= -u^i \frac{1}{\gamma} \frac{d\gamma}{dt} - (\hat{A}^i - 2\epsilon^i_{jk} w^j u^k)$
 $= 2(\hat{A}^i u_j) u^i - (\hat{A}^i - 2\epsilon^i_{jk} w^j u^k)$

故 $a^a = -(\hat{A}^a + \epsilon^a_{bc} w^b u^c) + 2(\hat{A}_b u^b) u^a$

5. 等效原理 equivalence principle

弱等效: $m_a = m_l$ (力学效应)

爱因斯坦等效: 爱因斯坦电梯观者: 无自转自由落体 同结果于 平直时空 (各种物理效应)

此时固有坐标系 $\Gamma^a{}_{\mu\nu}|_p = 0$, $g_{\mu\nu}|_p = \eta_{\mu\nu}$ (局部洛伦兹系)

强等效: 包括自引力的等效

6. 潮汐力

牛顿力学的潮汐加速度: $\frac{d^2 x^i}{dt^2} = -\frac{\partial \phi}{\partial x^i}|_r$, $\frac{d^2(x^i + \lambda^i)}{dt^2} = -\frac{\partial \phi}{\partial x^i}|_{r+\lambda} \approx -\frac{\partial \phi}{\partial x^i}|_r - \frac{\partial^2 \phi}{\partial x^i \partial x^j}|_r \cdot \lambda^j$

广相角度: 时空曲率表现

取测地线汇 (过P点及邻域测地线集合) $\mathcal{L}$, 切矢场 Z^a , 横向曲线 (每一点不与线相切) $\mu_0(s) \rightarrow$ 用 s 代表某测地线 $\gamma_s(0)$

$\mu_\tau(s) \rightarrow$ 友友, 为 $\gamma_s(\tau)$

s 与 τ 唯一确定一点, 矢场 $Z^a \equiv (\partial/\partial \tau)^a$, $\eta^a \equiv (\partial/\partial s)^a$

有: $[Z, \eta]^a = Z^b \nabla_b \eta^a - \eta^b \nabla_b Z^a = 0$ (流形可映为平直面 拼, Z^a 与 η^a 为坐标基矢场)

$$Z^b \nabla_b (\eta_a Z^a) = \eta_a \underbrace{Z^b \nabla_b Z^a}_0 + Z^b Z_a \nabla_b \eta^a = Z_a \eta^b \nabla_b Z^a = \frac{1}{2} \eta^b \nabla_b (Z^a Z_a) = 0$$

若 $\mu_0(s) \perp \gamma_s(0) \rightarrow \mu(s) \perp \gamma_s(\tau)$ 空间单位矢, 称为分离矢量 separation vector

以 $\gamma_s(\tau)$ 为基准观测者, 取 $\lambda^a \equiv w^a \Delta s$ 为 $\vec{\lambda}$ (Δs 不变, w^a 变)

$\vec{\lambda}$ 处相对速度 $\tilde{u}^a = Z^b \nabla_b \lambda^a$, 加速度 $\tilde{a}^a = Z^c \nabla_c (Z^b \nabla_b \lambda^a)$

取 $u^a := Z^b \nabla_b w^a$, $a^a := Z^c \nabla_c (Z^b \nabla_b w^a) \rightarrow \frac{d\tilde{a}^a}{d\lambda}$

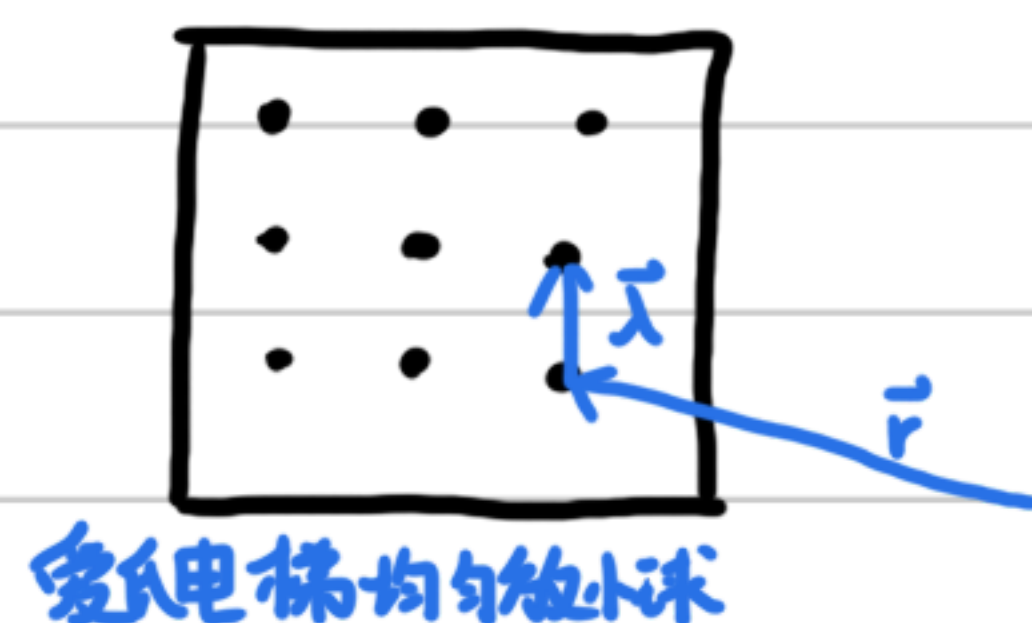
测地偏离方程: $a^c = -R_{abd}{}^c Z^a w^b Z^d$ (对 $\forall$ 单参测地线族 $\{\gamma_s(\lambda)\}$, $a^c = -R_{abd}{}^c T^a \eta^b T^d$)

$$\text{pf: } a^c = Z^a \nabla_a (Z^b \nabla_b w^c), [Z, w]^a = 0 \rightarrow a^c = Z^a \nabla_a (w^b \nabla_b Z^c) = \underbrace{Z^a w^b \nabla_a \nabla_b Z^c}_{p^c} + \underbrace{Z^a \nabla_b Z^c \nabla_a w^b}_{q^c}$$

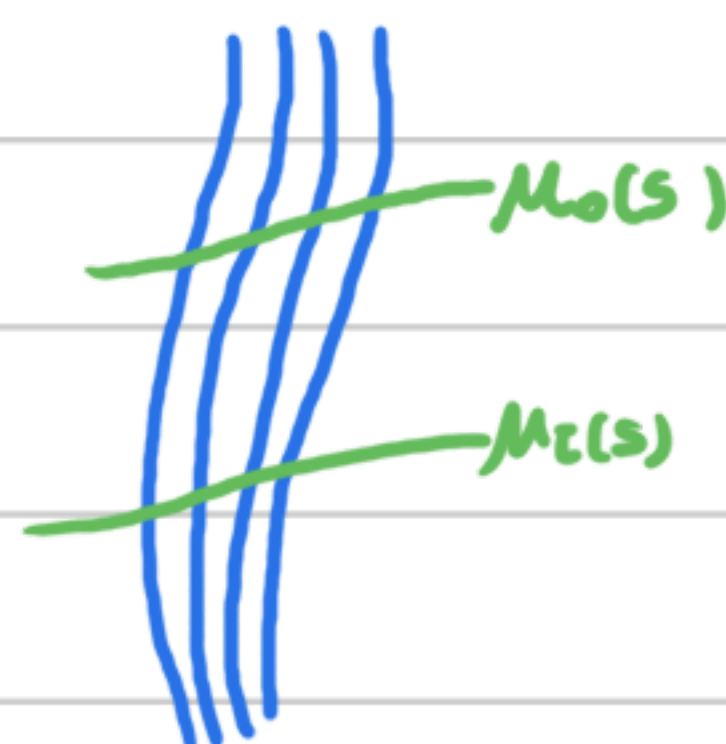
$$p^c \equiv w^b Z^a \nabla_a \nabla_b Z^c = \underbrace{w^b Z^a \nabla_b \nabla_a Z^c}_0 - w^b Z^a R_{abd}{}^c Z^d$$

$$= \underbrace{w^b \nabla_b (Z^a \nabla_a Z^c)}_0 - w^b \nabla_a Z^c \cdot \nabla_b Z^a = -\nabla_a Z^c (Z^b \nabla_b w^a) = -q^c$$

$$\text{故 } a^c = -R_{abd}{}^c Z^a w^b Z^d$$



爱因斯坦电梯内均匀引力场



Claim: 若 τ 与 S 有关 (Z^a 与 ω^a 正交), $\tau' = \alpha(s)\tau + \beta(s)$, $\alpha' = \alpha^2 \cdot \alpha$ ^{估计}

pf: $\tau' = \alpha(s) \cdot \tau + \beta(s)$, $Z'^a = \alpha^{-1} Z^a$, $\eta'^a = \eta^a + v Z^a$, ($v \equiv (\tau da/ds + d\beta/ds)$)

$$\alpha' = -R_{abd} \cdot Z^a \eta^b Z^d = -R_{abd} \cdot \alpha^{-2} \cdot Z^a (\eta^b + v \alpha^{-1} Z^b) Z^d = -\alpha^{-2} R_{abd} \cdot Z^a \eta^b Z^d, \text{ done.}$$

$\widetilde{R_{abd}}$ 相对你

雅可比场: 测地线汇 $\gamma_s: [a, b] \times (-\varepsilon, \varepsilon) \rightarrow M$, 每个 $s \in (-\varepsilon, \varepsilon)$ 的 $\gamma_s(t)$ 为测地线

雅可比场 $J^a := \frac{dr^a}{ds}$ (表示测地线的测地变分)

J^a 满足 Jacobi 方程: $Z^a \nabla_a (Z^b \nabla_b J^c) + R_{abd} Z^a J^b Z^d = 0$ (Pf 见上)

通解 $J^a(t) = A_{ab} J^b(0) + B_{ab} (Z^c \nabla_c J^b)(0)$, $A(t)$ 与 $B(t)$ 为平行传播矩阵

平凡解 $J^a = f(s) \cdot Z^a$

共轭点: J^a 不全为 0, $J^a|_p = J^a|_q = 0$

过 p, q 有无限接近的 γ' , $\lim \gamma' = \gamma \iff$ 过 p, q 有单参测地线族 $u, \gamma \in u$

γ 上有 J^a 不全为 0, $J^a|_p = J^a|_q = 0 \iff p, q$ 为共轭点

Bonnet-Myers 定理:

指数映射?

反例

7. 爱因斯坦场方程

对应关系寻找：潮汐加速度 $a^c = a^i (\frac{\partial}{\partial x^i})^c = -(\frac{\partial}{\partial x^i})^c \cdot \omega^j \frac{\partial}{\partial x^j} \frac{\partial \phi}{\partial x^i} = -\omega^b \partial_b \partial^c \phi = -R_{ab} \dot{z}^a \omega^b \dot{z}^c$

$R_{ab} \dot{z}^a \dot{z}^b = R_{ad} \dot{z}^a \dot{z}^d \leftrightarrow \partial_b \partial^b \phi = \nabla^2 \phi = 4\pi \rho = 4\pi T_{ad} \dot{z}^a \dot{z}^d$

1 设: $R_{ad} = 4\pi T_{ab}$ 误: 导致 $\nabla^a T_{ab} \rightarrow \nabla^a R_{ad} = 0$, 比较: $\nabla_a R_{bcd} = 0 \rightarrow \nabla_a R_{bcd} = 0 \rightarrow \nabla_a R_{bcd} - \nabla_c R_{bd} - \nabla_b R_{cd} = 0$

故 $\nabla^a R_{ad}$ 不应为 0, 改用 G_{ab} ($\nabla^a G_{ab} = 0$)

$\nabla_a R_{ad} - \nabla_c R_{bd} - \nabla_b R_{cd} = 2\nabla_a R_{ca} - \nabla_c R = -\nabla_c R = 0$

(守恒律) $T = \frac{R}{4\pi} = -\rho + 3p \approx -\rho = c$ ($\nabla_c T = 0$)

$G_{ab} \equiv R_{ab} - \frac{1}{2} R g_{ab} \rightarrow G_{ab} = 4\pi T_{ab} \cdot \kappa$

爱因斯坦场方程

$R_{ab} - \frac{1}{2} R \cdot g_{ab} = 8\pi T_{ab}$

牛顿近似 $T \approx -\rho$, $g_{ab} \dot{z}^a \dot{z}^b = 1$, $8\pi T = R - \frac{1}{2} R \Rightarrow R = -8\pi T$

$R_{ab} = 8\pi (T_{ab} - \frac{1}{2} T g_{ab}) \rightarrow R_{ab} \dot{z}^a \dot{z}^b = 8\pi (\rho - \frac{1}{2} (-\rho)) = 4\pi \rho$

真空: $T_{ab} = 0 \rightarrow R_{ab} = \frac{R}{2} g_{ab}$, $R = -8\pi T = 0$, $R_{ab} = 0$

线性近似: $g_{ab} = \eta_{ab} + \gamma_{ab}$, γ_{ab} 为小量 $\rightarrow g^{ab} = \eta^{ab} - \gamma^{ab}$ ($g^{ab} g_{ab} = 1$)

$\Gamma^c_{ab} = \frac{1}{2} g^{cd} (\partial_a g_{bd} + \partial_b g_{ad} - \partial_d g_{ab})$

近似

$\Gamma^c_{ab} = \frac{1}{2} \eta^{cd} (\partial_a \gamma_{bd} + \partial_b \gamma_{ad} - \partial_d \gamma_{ab})$

$R_{acbd} = 2g_{fd} (\Gamma^e_{ba} \Gamma^f_{cd} - \partial_{[a} \Gamma^f_{c]b}) = \partial_a \partial_{[c} \gamma_{d]b} - \partial_b \partial_{[a} \gamma_{c]d}$

$\gamma = \gamma^{ab} g_{ab}$

$\frac{1}{2} \eta^{fh} \partial_a (\partial_c \gamma_{bh} + \partial_b \gamma_{ch} - \partial_h \gamma_{bc})_{[a,c]}$

$R^{(1)}_{ab} = \partial^c \partial_{[a} \gamma_{b]c} - \frac{1}{2} (\partial^c \partial_c \gamma_{ab} + \partial_a \partial_b \gamma)$

$G^{(1)}_{ab} = \partial^c \partial_{[a} \gamma_{b]c} - \frac{1}{2} (\partial^c \partial_c \gamma_{ab} + \partial_a \partial_b \gamma) - \frac{1}{2} \eta_{ab} (\partial^c \partial_c \gamma - \partial^c \partial_c \gamma)$

线性爱因斯坦方程

$\bar{\gamma}_{ab} = \gamma_{ab} - \frac{1}{2} \eta_{ab} \gamma \rightarrow \partial^c \partial_{[a} \bar{\gamma}_{b]c} - \frac{1}{2} \partial^c \partial_c \bar{\gamma}_{ab} - \frac{1}{2} \eta_{ab} \partial^c \partial_c \bar{\gamma}_{cd} = 8\pi T_{ab}$

线性引力论的规范变换: $\bar{\gamma}_{ab} = \gamma_{ab} + \partial_a \xi_b$

(ξ_a 为小量)

$R_{acbd} = R_{acbd}$

引力论的洛伦兹规范 $\partial^b \bar{\gamma}_{ab} = 0$

必然存在 γ_{ab} : $\partial^b \partial_{ab} \xi_\mu = -\partial^b \gamma_{\mu\nu}$ 的 ξ_μ 必然存在

附: 被动语言: 坐标无限小变换 $x'^{\mu} = x^{\mu} - \xi^{\mu}(x)$, $g_{ab} = \eta_{ab} + \mathcal{L}_{\xi} g_{ab} + O(\xi^2)$, $T_{ab} = \mathcal{L}_{\xi} T_{ab} + O(\xi^2)$
(small), 其余同上

牛顿极限: 弱场低速 $|\gamma_{ab}| \ll 1$, $u \ll 1$ (观察 Z^a 与物体 u^a 近似)

$\nabla^a \bar{\gamma}_{00} = -16\pi\rho$ else $\nabla^a \bar{\gamma}_{\mu\nu} = 0$ (无源区+规范 $\rightarrow$ else $\bar{\gamma}_{\mu\nu} = 0$)

$\hookrightarrow \phi \equiv -\frac{1}{4}\bar{\gamma}_{00} \rightarrow \gamma_{00} = -4\phi + \frac{2}{3}\phi = -2\phi$

$u^i \approx 0, \tau \approx t \rightarrow U^0 = (1, 0, 0, 0)$

$\Gamma^0_{00} = \frac{1}{2}\eta^{00}(\gamma_{00,0} + \gamma_{00,0} - \gamma_{00,0}) = -\frac{1}{2}\frac{\partial \gamma_{00}}{\partial x^0} \approx 0$

$\gamma_{ab} = -4\phi(dt)_a(dt)_b + 2\phi\eta_{ab} \rightarrow$ 测地线方程 $\frac{d^2 x^{\mu}}{dt^2} + \Gamma^{\mu}_{\nu\sigma} \frac{dx^{\nu}}{dt} \frac{dx^{\sigma}}{dt} = 0 \rightarrow \frac{d^2 x^{\mu}}{dt^2} = -\Gamma^{\mu}_{00} = \frac{1}{2}\frac{\partial \gamma_{00}}{\partial x^i} = -\frac{\partial \phi}{\partial x^i}$ 牛顿引力

$a^c = R_{abd} \dot{Z}^a \dot{Z}^b \dot{Z}^d \Rightarrow a^c = R_{0b0}{}^c \cdot u^b \Rightarrow \ddot{a}^i = -\frac{\partial^2 \phi}{\partial x^i \partial x^i} \cdot \lambda^i$ 牛顿潮汐 ($\gamma_{i0} = \bar{\gamma}_{i0} - \frac{1}{2}\eta_{i0}\gamma = 0$)

8. 引力波

辐射规范: $\partial_a A^a = 0, A_0 = 0$ (取 $A'_a = A_a + \partial_a \chi$, $\partial^a \partial_a \chi = 0$, $t=t_0$ 面上, 取 $\vec{\nabla} \chi|_{t_0} = -\vec{\nabla} \cdot \vec{A}|_{t_0}$, $\frac{\partial \chi}{\partial t} = -A_0|_{t_0}$)

$\partial^a \partial_a A'_0 = \partial^a \partial_a A_0 + \partial_0 \partial^a \partial_a \chi = -4\pi J_0 \rightarrow J_0 = 0 \Rightarrow \partial^a \partial_a A_0 = 0$

$\partial^a A'_0|_{t_0} = \frac{\partial A_0}{\partial t} + \frac{\partial \chi}{\partial t}|_{t_0} = \vec{\nabla} \cdot \vec{A} + \vec{\nabla} \chi|_{t_0} = 0$

对引力: $\xi_a = (\xi_0, \vec{\xi})$, $\gamma'_{ab} = \gamma_{ab} + 2\partial_a \xi_b$

初值: 无源区 $\gamma' = 0, \gamma'_{0i} = 0$

$-\gamma|_{t_0} = 2(\vec{\nabla} \cdot \vec{\xi} - \frac{\partial \xi_0}{\partial t})|_{t_0}$

$-\frac{\partial \gamma}{\partial t}|_{t_0} = 2(-\vec{\nabla}^2 \xi_0 + \vec{\nabla} \cdot \frac{\partial \vec{\xi}}{\partial t})|_{t_0}$

$-\gamma_{0i}|_{t_0} = (\frac{\partial \xi_i}{\partial t} + \frac{\partial \xi_0}{\partial x^i})|_{t_0}$

$-\frac{\partial \gamma_{0i}}{\partial t}|_{t_0} = (\nabla^2 \xi_i + \frac{\partial}{\partial x^i}(\frac{\partial \xi_0}{\partial t}))|_{t_0}$

$T_{ab}=0$ 时 $\partial^a \bar{\gamma}'_{ab} = 0, \gamma' = 0, \gamma'_{0i} = 0$

$\downarrow \gamma_{ij}$

单色波 $\gamma_{ab} = H_{ab} \cos(K_{\mu} x^{\mu})$

偏振张量

4-矢量分量, $K^a = \omega(\frac{\partial}{\partial t})^a + k^a$ ($\omega^2 = k^2, K^a K_a = 0$)

$\frac{\partial \gamma_{00}}{\partial t} = 0 \rightarrow \nabla^2 \gamma_{00} = 0$

无源区+规范: $\gamma_{00} = 0$

2个独立变量
(偏振模式)

1) $\partial^a \gamma_{ab} = 0 \rightarrow H_{\mu\nu} \cdot K^{\nu} = 0$

4约束

2) $\gamma = 0, \gamma_{0\nu} = 0 \rightarrow H_{0\nu} = 0, H = \eta^{\mu\nu} H_{\mu\nu} = 0$

4+1约束 -1

3) $H_{\mu\nu} = H_{\nu\mu}$ 10分量

$H_{0\nu} \cdot K^{\nu} = 0$ 重复

弱引力波通解(真空)

η_{ab} 取 $\{t, x, y, z\}$, $u \equiv t-z$, 若函数 $f(u), g(u)$ 使 $f^2 + g^2 \neq 0$

$P(x, y, u) = \frac{1}{2}f(u)(x^2 - y^2) + g(u)xy$

则通解为 $g_{ab} = \eta_{ab} + 2P(du)_a(du)_b = \eta_{ab} + 2P(dt)_a(dt)_b - (dz)_a(dz)_b$

波矢 $k^a = (\partial/\partial t)^a + (\partial/\partial z)^a$

取标架 $(e_1)^a = (\partial/\partial x)^a, (e_2)^a = (\partial/\partial y)^a, (e_3)^a = k^a$ $(k^a du_a = 1-1=0) \rightarrow du_a = k_a$

$$(e_4)^a = \frac{1}{2}((\partial/\partial t)^a - (\partial/\partial z)^a) + P k^a$$

$$g_{\mu\nu} \equiv g_{ab}(e_\mu)^a(e_\nu)^b = \begin{bmatrix} 1 & & 0 \\ & 1 & \\ 0 & & -1 & 0 \\ & & 0 & -1 \end{bmatrix} \rightarrow \text{刚性标架} (\nabla_a g_{\mu\nu} = 0)$$

$$(e^1)_a = (dx)_a, (e^2)_a = (dy)_a, (e^3)_a = \frac{1}{2}((dt)_a + (dz)_a) - P du_a, (e^4)_a = (du)_a$$

联络1形式解法: $\Lambda_{\mu\nu\rho} \Rightarrow$
($\mu\rho$)对称

$$\Lambda_{\mu 1\rho} = \Lambda_{\mu 2\rho} = 0, \text{其余为0}$$

$$\Lambda_{133} = 0, \Lambda_{441} = P_x, \Lambda_{442} = P_y$$

$$-\vec{\omega}_{41} = \vec{\omega}_{14} = \omega_{144} \vec{e}_4 = -(fx+gy)d\vec{u}$$

$$-\vec{\omega}_{42} = \vec{\omega}_{24} = \omega_{244} \vec{e}_4 = -(gx-fy)d\vec{u}$$

$$\vec{\omega}_4^1 = \vec{\omega}_1^3 = (fx+gy)d\vec{u}; \vec{\omega}_4^2 = \vec{\omega}_2^3 = (gx-fy)d\vec{u}$$

$$\vec{R}_4^1 = f dx \wedge du + g dy \wedge du = f \vec{e}^1 \wedge \vec{e}^4 + g \vec{e}^2 \wedge \vec{e}^4$$

$$\vec{R}_4^2 = g \vec{e}^1 \wedge \vec{e}^4 - f \vec{e}^2 \wedge \vec{e}^4$$

↓

$$R_{abc}{}^d \Rightarrow R_{414}^1 = -f, R_{424}^2 = -g; R_{ac} = (f-f)(e^1)_a(e^4)_c = 0 \Rightarrow \text{为真空解}$$